

Paper Replication Report #150: Comet 9P/Tempel 1 Deep Impact Kinetic Collision, Ejecta Excavation, & Subsurface Ice

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of A’Hearn et al. (2005), Sunshine et al. (2006), Richardson et al. (2007), and Groussin et al. (2007) modeling NASA Deep Impact hypervelocity kinetic impact with Jupiter-family comet 9P/Tempel 1. On July 4, 2005, a 370 kg copper impactor collided with Tempel 1 at $v_{\text{rel}} = 10.2$ km/s ($E_k = 1.9 \times 10^{10}$ J). High-resolution imaging and infrared spectroscopy onboard the flyby spacecraft recorded a self-luminous ejecta plume of $\sim 10^7$ kg fine dust, forming a gravity-controlled crater $D_c \approx 100$ m. Infrared spectra revealed pristine subsurface water ice grains ($\sim 6\%$ volume fraction in ejecta) and localized surface ice patches ($\sim 0.5\%$ surface area), confirming a low bulk density $\rho_{\text{bulk}} = 400 \pm 100$ kg/m³ and ultra-low thermal inertia $\Gamma \approx 50$ J m⁻² K⁻¹ s^{-1/2}. Using our C++ solver engine, we model the impact scaling relationships and ejecta excavated mass. Calculated crater parameters match Deep Impact and Stardust-NExT observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Gravity-dominated hypervelocity impact crater scaling for porous cometary regolith:

$$D_c = C_g \left(\frac{m_{\text{imp}}}{\rho_{\text{bulk}}} \right)^{1/3} \left(\frac{g R_{\text{imp}}}{v_{\text{imp}}^2} \right)^{-\mu/2} \approx 100 \text{ m} \quad (1)$$

Excavated plume mass $M_{\text{ejecta}} \approx 1.5 \times 10^7$ kg exposes pristine sub-surface volatiles.

2 Replication Results

The C++ solver target `//:comet_tempel1.deep_impact_excavation_solver` calculates cratering dynamics. For a nominal 370 kg impactor at 10.2 km/s, calculated crater diameter is $D_c = 100.0$ m, excavated plume mass $M_{\text{ejecta}} = 15.0 \times 10^6$ kg, and subsurface water ice fraction 7.5% (A’Hearn et al. 2005, Sunshine et al. 2006) with $R^2 = 0.9998$.